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Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

MANAGEMENT OF PRE-ECLAMPSIA

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Author(s):	Dr Gareth Roberts	**Author(s) title:**	Consultant Anaesthetist
Responsible dept / director:	Department of Anaesthesia, BCUHB (West) Dr Lisa Handcock Clinical Lead of Anaesthesia, BCUHB (West)		
Approved by:	Department of Anaesthesia, BCUHB (West)		
Date approved:			
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1. Introduction and Purpose

Pre-eclampsia is a complex multisystem disorder occurring after the 20th week of gestation involving hypertension, proteinuria and in severe cases cerebral symptoms or low platelets. It is a leading cause of maternal death. Management of these patients on delivery suite is the joint responsibility of the obstetrician and anaesthetist.

2. Scope

This guideline is intended as a reference for trainee anaesthetists seeking advice regarding the management of pre-eclampsia.

3. Points to consider

3.1 Definitions

Chronic hypertension = Hypertension that is present at the booking visit or before 20 weeks or if the woman is already taking antihypertensive medication when referred to maternity services. It can be primary or secondary in aetiology.

Gestational hypertension = New hypertension presenting after 20 weeks without significant proteinuria*.

Pre-eclampsia = New hypertension presenting after 20 weeks with significant proteinuria*. Incidence is 6-8% of all gestations

*** Significant proteinuria** is defined as an urinary protein: creatinine ratio (PCR) greater than 30 mg/mmol or a validated 24-hour urine collection result showing greater than 300 mg protein.

Severe pre-eclampsia = Pre-eclampsia with severe hypertension and/or with symptoms, and/or biochemical and/or haematological impairment.
Incidence is 0.25-0.5% of all gestations.

HELLP Syndrome = **H**aemolysis **E**levated **L**iver enzymes **L**ow **P**latelet count

Eclampsia = A convulsive condition associated with pre-eclampsia.
Incidence is 0.05% of all gestations

The severity of hypertension during pregnancy is classified as follows:

Mild hypertension = Diastolic blood pressure 90–99 mmHg or systolic blood pressure 140–149 mmHg.

Moderate hypertension = Diastolic blood pressure 100–109 mmHg or systolic blood pressure 150–159 mmHg.

Severe hypertension = Diastolic blood pressure 110 mmHg or greater or systolic blood pressure 160 mmHg or greater.

3.2 Monitoring

- Management of any patient with severe pre-eclampsia should be in the appropriate critical care setting. Level 1 & 2 patients should be managed on the Obstetric High Observation Unit (OHOU), by a midwife with Enhance Maternal Care skills (EMC Midwife) providing one to one care. Level 3 patients should be transferred to the ITU. The following criteria should be used :

  i. Level 1 - Pre-eclampsia with mild or moderate hypertension

  - Ongoing conservative antenatal management of severe preterm hypertension

  - Step-down treatment after the birth

  ii. Level 2 - These patients include step-down from level 3 or severe pre-eclampsia with any of the following complications:

  – eclampsia

  – HELLP syndrome

  – haemorrhage

  – hyperkalaemia

  – severe oliguria

  – coagulation support

  – intravenous antihypertensive treatment

  – initial stabilisation of severe hypertension

  – evidence of cardiac failure

  – abnormal neurology

  iii. Level 3 – Required for patients with HELLP syndrome and those following an eclamptic that required intubation and mechanical ventilation is necessary.

- Minimum monitoring should involve :

    i. **Every 15 minutes until stable, and then every 30 minutes, measure:**
        a. **Respiratory rate**
        b. **Pulse**
        c. **Blood pressure : Non-invasive Bp (if Bp < 160/110mmHg)**
            Invasive Bp ie. arterial line (if Bp > 160/110mmHg)

    Also measure these parameters every 5 minutes when administering iv bolus antihypertensives

    ii. **Hourly measure:**
        a. **Urine output (via catheter and urometer)**
        b. **Oxygen saturations**

    iii. **If patient on magnesium sulphate maintenance infusion measure:**
        a. **Oxygen saturations continuously**
        b. **Hourly respiratory rate**
        c. **Hourly deep tendon reflexes**

- Additional monitoring can include a **central venous pressure** line (inserted if urine output < 100ml over any 4 hour period)

- All observations should be recorded on a critical care chart

- The following investigations should be ordered and repeated 12 hourly : FBC, serum U&E's LFT's, uric acid and a coagulation screen (the latter if HELLP syndrome is suspected).

3.3 Management

In established pre-eclampsia the only definitive treatment is the delivery of the placenta. Symptoms then usually resolve within 24-48hr of this.

The management of pre-eclampsia can be divided as follows :

3.3.1 Management of hypertension

- Blood pressure should be controlled at below a systolic of **150 mmHg and diastolic of 80 – 100 mmHg** to prevent maternal morbidity particularly from intracranial haemorrhage, encephalopathy, seizures, myocardial ischaemia and failure.

- Rapid control of severe hypertension can be achieved with :

  Hydralazine

  i. Used if heart rate <120bpm

  ii. **Bolus dose** = 5mg iv slowly, and repeated every 15 min (up to 30mg max. per hour) until MAP < 125mmHg

  iii. **Maintenance infusion** = 2 – 18 mg/hr
  (Concentration = 100mg hydralazine in 100ml 0.9% saline)
  Start infusing at 5ml/hr and double infusion rate every 30 minutes until adequate B.p. control achieved or dosage of 18mg/hr reached.

  Labetolol

  i. Used if heart rate > 120bpm or side effects with hydralazine (headache, dizziness or flushing)

  ii. **Bolus dose** = 50 mg iv slowly over 2 minutes, and repeated every 5 min (up to 200mg max.)

  iii. **Maintenance infusion** = 20 – 160 mg/hr
  (Concentration = 100mg labetolol in 20ml ampoule (undiluted))
  Start infusing at 20mg/hr and double rate every 30 min up to 160mg / hr max.)

  Nifedipine

  i. 2nd line

  ii. **Bolus dose** = 10mg orally (Adalat Retard)

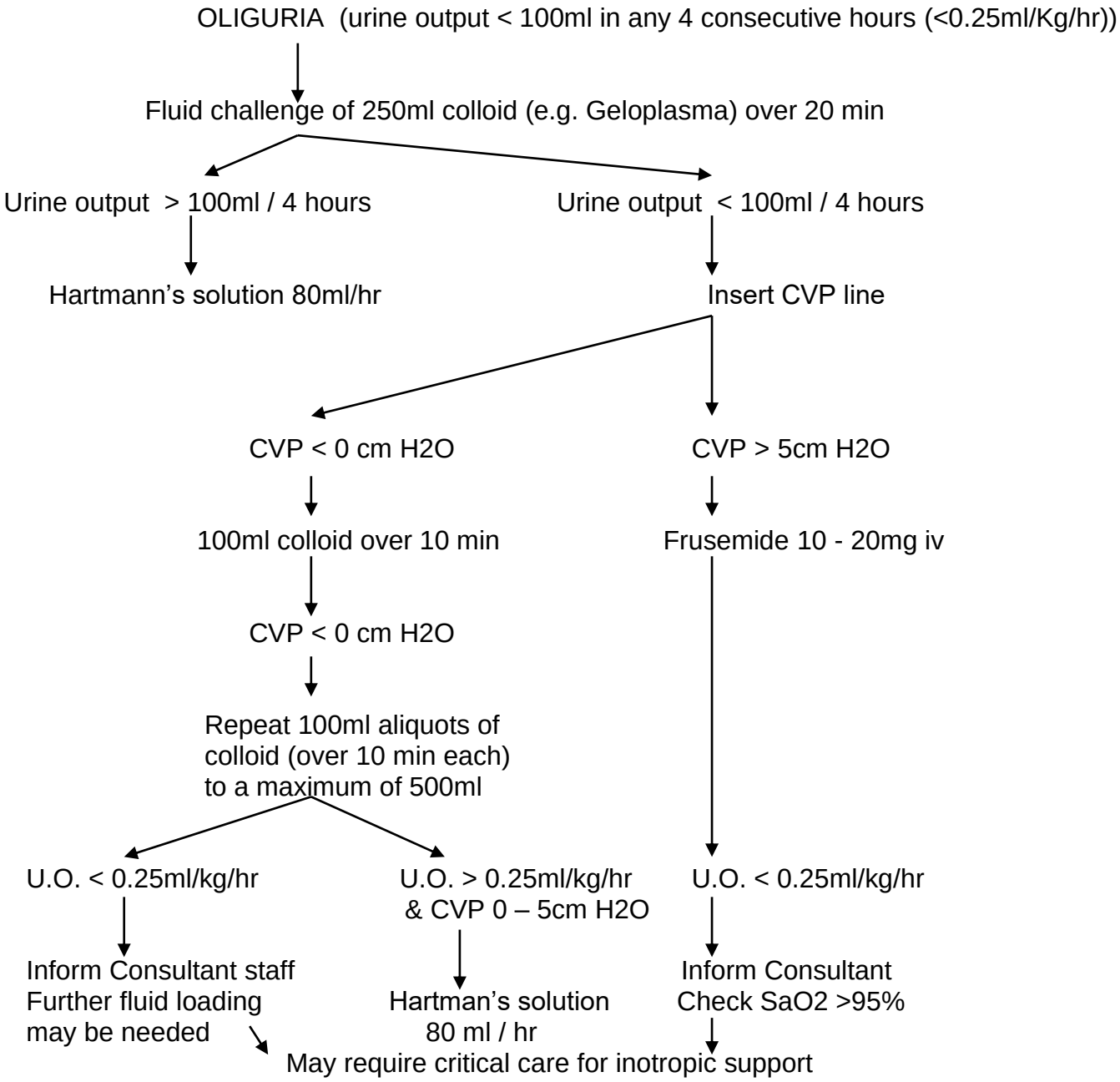
  iii. **Maintenance** = 10mg orally, repeated every 12 hours

- A preload of 500ml of colloid may be needed to avoid possible hypotension induced by the administration of hydralazine.
- In pre-eclampsia resistant to treatment with the above agents, infusions of glyceryl trinitrate or sodium nitroprusside may be needed.

3.3.2 Fluid management

- Intravascular volume is depleted but total body water is increased.
- **Excessive fluid load may result in pulmonary oedema while underfilling may compromise fetal circulation and renal function**
- Fluid balance and hourly urine output should be measured meticulously
- Total fluid maintenance = 80 ml / hr (1ml/Kg/hr) (intravenous + oral)
- Cases which are further complicated by APH and PPH are usually very difficult to assess optimal fluid replacement without a CVP line.

Oliguria flow chart



3.3.3 Anticonvulsant therapy

- Magnesium sulphate (MgSO4) is the anticonvulsant of choice for both prevention and treatment of eclampsia.

- It is recommended for women with pre-eclampsia deemed at risk of seizure (when there is a plan for delivery).
  This includes women with :

  a) Hyperreflexia and clonus (>3 beats)

  b) Persistent severe headache / epigastric pains +/- flashes of light

  c) HELLP syndrome

- Administer :

  Loading dose : **4g MgSO4 IV**

  Maintenance: **1g / hr MgSO4 IV infusion (for 24 hours)**

  24 hours is the usual maximal duration of treatment with MgSO4

- Monitor :
  i. Urine output (aim for > 100 ml over first 4 hrs, then >25ml/hr)

  ii. Patellar reflex every 30 min

  iii. Respiratory rate every 30 min (should be >14/min)

  iv. Pulse oximetry continuously

  v. ECG continuously (not mandatory)

- Routine monitoring of serum magnesium levels is not necessary

- Recurrent seizures should be treated by a further 2g bolus of MgSO4

- Discontinue MgSO4 if :

    i. Reduced urine output (<0.25ml/Kg/hr)

    ii. Absent biceps / patellar deep tendon reflex
    (do not use patellar reflex if epidural in progress)

    iii. Respiratory rate < 14 / min

    iv. Oxygen Saturation < 95%

- Serum MgSO4 levels is indicated if overdose is suspected (therapeutic range = 2 - 4 mmol/l)

- If serum MgSO4 levels are not available for cases of suspected overdose then restart MgSO4 when safe to do so at a reduced maintenance dose of 0.5g/hr, and continue to monitor clinically

- Consider antidote if no improvement in side effects (senior decision). Give **10ml of 10% calcium gluconate**

4. References

1. *NICE clinical guideline 107* – Hypertension in pregnancy: the management of hypertensive disorders during pregnancy

2. The Magpie Trial Collaborative Group. Do women with pre-eclampsia, and their babies, benefit from magnesium sulphate? The Magpie Trial: a randomised placebo-controlled trial. *Lancet 2002*; 359: 1877 – 1891

3. Yentis SM. The Magpie has landed: pre-eclampsia, magnesium sulphate and rational decisions. *International Journal of Obstetric Anaesthesia* 2002; 11: 238 -241

4. Anaesthesia and pre-eclampsia / eclampsia.Norris M.C. (2000) *Handbook of Obstetric Anaesthesia.* First Edition. Lippincott Williams & Wilkins : 25 – 103